

Sphere-ICPO: A Spherical Vector-Based Improved Crested Porcupine Optimizer for UAV 3D Path Planning

Abstract

The Crested Porcupine Optimizer (CPO) is a recently proposed metaheuristic with strong performance on global optimization benchmarks. However, existing CPO improvements for UAV path planning all operate in Cartesian coordinates, neglecting the advantages of spherical vector encoding pioneered by the SPSO framework. When naively transplanted into spherical vector space, CPO exhibits three structural weaknesses: ineffective exploration due to random individual differentiation in a highly constrained search space, a rigid random exploration–exploitation switch, and the absence of a local-optimum escape mechanism. This paper proposes Sphere-ICPO, which combines SOS mutualism, a nonlinear exploration schedule, and stagnation-triggered retreat in spherical vector space. On four preregistered LiDAR DEM scenarios, all five algorithms were evaluated with 500 particles, 200 iterations, and 30 independent runs. Sphere-ICPO obtained the lowest mean cost on three maps and the best average rank (1.50); the Friedman test detected an overall difference ($p = 0.00869$). Holm-corrected pairwise tests show significant gains over all baselines on Map 2 and over CPO and WOA on every map. Ablation results identify SOS mutualism as the robust core component, while the adaptive schedule and retreat do not independently reduce mean cost on the selected ablation scenario.

1 Introduction

Three-dimensional (3D) path planning is a fundamental enabler for autonomous unmanned aerial vehicles (UAVs), with applications spanning logistics delivery, infrastructure inspection, environmental monitoring, and emergency response [5, 6, 7]. The problem requires generating a collision-free trajectory from a start point to a destination while simultaneously optimizing path length, safety, altitude, and smoothness under nonlinear constraints imposed by terrain, no-fly zones, and UAV kinematics. Metaheuristic algorithms have become the dominant solution paradigm due to their gradient-free nature and flexibility in handling multi-objective optimization [1, 4].

Among recently proposed metaheuristics, the Crested Porcupine Optimizer (CPO) [4] stands out for its rich behavioral repertoire. Inspired by four defensive strategies of crested porcupines—visual intimidation, acoustic deterrence, olfactory repellent, and physical attack—CPO organizes search into distinct exploration (visual, acoustic) and exploitation (olfactory, physical) phases. CPO has demonstrated competitive performance on CEC benchmark suites and has rapidly spawned improvement efforts. Liu et al. [5] proposed ICPO, integrating Symbiotic Organisms Search (SOS) mutualism and periodic retreat, achieving a 63% cost reduction on urban terrain. Liu et al. [6] developed QCPO with Q-learning for adaptive parameter control, Zhang et al. [7] proposed JAYA-CPO fusing JAYA with CPO’s attack strategy, Zhang et al. [8] introduced Multi-ICPO with Tent chaotic mapping and Cauchy-Gaussian mutation, and Shi and Li [9] extended CPO to multi-objective optimization as MOCPO.

A critical observation motivates this work: **all existing CPO variants operate in Cartesian coordinate space**, where waypoints are encoded directly as (x, y, z) coordinates. Cartesian encoding bears no intrinsic relationship to UAV motion parameters, requiring kinematic constraints to be enforced indirectly through penalty functions. An alternative paradigm—spherical vector encoding—was introduced by Phung and Ha [1] in their Spherical Vector-based PSO (SPSO). Each path segment is represented as (ρ, ψ, ϕ) , mapping directly to the UAV’s speed, climbing angle, and turning angle, thereby encoding kinematic feasibility into the search bounds and transforming the search into the UAV’s configuration space. **No prior work** has investigated CPO under spherical vector encoding, leaving open the question of whether CPO’s defense strategies can be effectively adapted to this representation.

Our preliminary experiments reveal that naïvely transplanting CPO into the spherical vector framework produces poor performance. We identify three structural root causes:

1. **Ineffective exploration through random differentiation.** CPO’s visual and acoustic strategies generate search directions via differences between randomly selected individuals $(\mathbf{x}_{r1} - \mathbf{x}_{r2})$. In spherical space, where constraint violations produce infinite cost, randomly selected individuals are frequently infeasible, rendering the differential signal meaningless.
2. **Rigid exploration–exploitation switching.** CPO employs $\text{rand}() < \text{rand}()$ to choose between exploration and exploitation, yielding a fixed 50% probability regardless of the optimization stage. This prevents the natural transition from broad initial exploration to focused late-stage exploitation.
3. **Absence of local-optimum escape.** Once the population converges

to a suboptimal basin, none of CPO’s four strategies can redirect the search. The olfactory and physical attack strategies refine locally, while the visual and acoustic strategies lack the perturbation magnitude to cross basin boundaries.

To address these three limitations, this paper proposes **Sphere-ICPO**, which adapts and extends CPO for spherical vector-based UAV path planning. Our contributions are:

1. **SOS mutualism in spherical space.** We replace CPO’s visual and acoustic strategies with SOS mutualism, where each agent cooperates with a randomly selected partner and is jointly guided toward the global best. Unlike random differentiation, this mechanism provides robust directional guidance regardless of partner feasibility. This is the first application of SOS mutualism under spherical vector encoding.
2. **Adaptive exploration ratio.** We replace $\text{rand}() < \text{rand}()$ with a nonlinear decay function $a(t) = 2(0.7(1 - t/T)^{0.5} + 0.3)$, enabling a smooth transition from $\approx 100\%$ exploration to $\approx 30\%$ exploitation over the course of optimization.
3. **Stagnation-triggered retreat.** Unlike prior work [5] that triggers population retreat at fixed intervals—potentially disrupting productive convergence—Sphere-ICPO activates retreat only when the global best stagnates for five consecutive iterations. Retreat uses cosine-geometric back-stepping with logarithmically decaying step size, providing targeted escape without unnecessary perturbation.

Comprehensive experiments on four frozen benchmark scenarios constructed from real LiDAR DEM data—spanning 4 to 7 threats—show that Sphere-ICPO attains the lowest mean cost on three maps and the best average rank among five algorithms. The ablation study confirms SOS mutualism as the robust core improvement, but does not support independent mean-cost gains from every auxiliary component. Holm-corrected Wilcoxon tests and a Friedman test ($p = 0.00869$) provide a more conservative statistical assessment.

The remainder of this paper is organized as follows. Section 2 formulates the path planning problem and the spherical vector encoding framework adopted from SPSO [1]. Section 3 details the proposed Sphere-ICPO algorithm. Section 4 reports experimental results including ablation, comparison, and statistical analyses. Section 5 concludes the paper.

2 Problem Formulation

2.1 Spherical Vector Encoding

Following the SPSO framework, a UAV flight path is encoded as a set of N spherical vectors:

$$\Omega = (\rho_1, \psi_1, \phi_1, \rho_2, \psi_2, \phi_2, \dots, \rho_N, \psi_N, \phi_N), \quad N = n - 2 \quad (1)$$

where n is the total number of waypoints (including fixed start and end points). Each triplet (ρ_j, ψ_j, ϕ_j) represents the displacement from waypoint j to $j + 1$, with magnitude $\rho_j \in [0, \text{path_length}]$, elevation angle $\psi_j \in [-\pi/4, \pi/4]$, and azimuth angle $\phi_j \in [\phi_0 - \pi/4, \phi_0 + \pi/4]$ where ϕ_0 is the direction to the destination.

The mapping from spherical vectors to Cartesian waypoints is:

$$x_{j+1} = x_j + \rho_j \cos \psi_j \sin \phi_j \quad (2)$$

$$y_{j+1} = y_j + \rho_j \cos \psi_j \cos \phi_j \quad (3)$$

$$z_{j+1} = z_j + \rho_j \sin \psi_j \quad (4)$$

where (x_1, y_1, z_1) and (x_n, y_n, z_n) are the fixed start and end points, respectively.

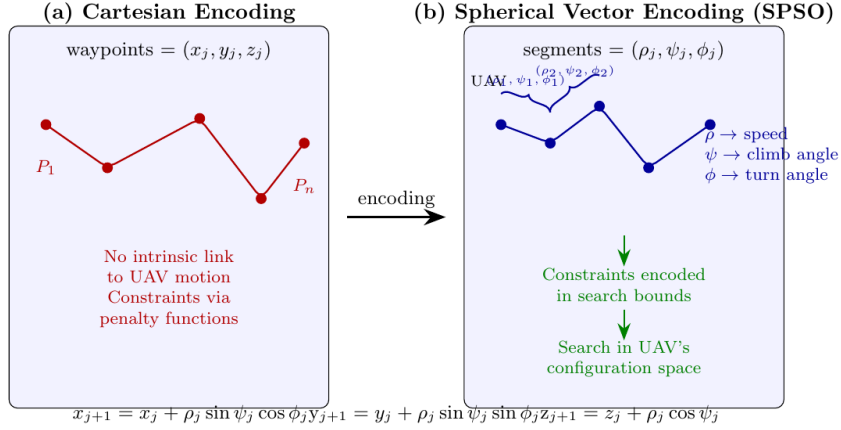


Figure 1: Spherical vector encoding: (a) Cartesian coordinates bear no intrinsic relationship to UAV motion, requiring penalty-based constraint enforcement; (b) spherical vectors (ρ, ψ, ϕ) map directly to the UAV's speed, climbing angle, and turning angle, encoding kinematic feasibility into the search bounds.

2.2 Cost Function

Following the SPSO cost model, the overall path cost $F(\Omega)$ is a weighted sum of four components:

$$F(\Omega) = b_1 F_1(\Omega) + b_2 F_2(\Omega) + b_3 F_3(\Omega) + b_4 F_4(\Omega) \quad (5)$$

where F_1 is the total path length (sum of Euclidean distances between consecutive waypoints), F_2 is the threat avoidance cost based on a cylindrical obstacle model with three-tier penalty (safe=0, warning=linear, collision= ∞), F_3 is the altitude cost penalizing deviations from the midpoint of the allowable altitude range $[h_{\min}, h_{\max}]$, and F_4 is the smoothness cost combining turning angle and climbing angle variations. The weights are set to $b_1 = 5$, $b_2 = 1$, $b_3 = 10$, $b_4 = 1$, as established by the SPSO framework.

3 Proposed Algorithm: Sphere-ICPO

3.1 Overview

Sphere-ICPO retains the four-defense-strategy framework of CPO while introducing three targeted innovations that address the specific challenges of spherical vector space optimization. Algorithm 1 presents the pseudo-code.

Algorithm 1: Sphere-ICPO

Input: Population size N , max iterations T , model

Output: Best path Ω^*

1. Initialize population with random spherical vectors
2. Evaluate initial fitness, identify global best $\mathbf{x}_{\text{best}}$
3. **for** $t = 1$ **to** T **do**
4. Compute adaptive exploration ratio: $a(t) = 2(0.7(1 - t/T)^{0.5} + 0.3)$
5. **for** each agent i **do**
6. **if** $\text{rand}() < a(t)/2$ **then**
7. *SOS Mutualism:* $\mathbf{RMV} = \mathbf{x}_i + r_1(\mathbf{x}_{\text{rand}} - \mathbf{x}_i)$
 $\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + r_2(\mathbf{x}_{\text{best}} - \mathbf{RMV})$
8. **else**
9. *CPO Exploitation:* Apply odor (Strategy 3) or physical attack (Strategy 4)
10. **end if**

11. Enforce bounds, evaluate fitness, update $\mathbf{x}_{\text{best}}$
12. **end for**
13. **if** $\mathbf{x}_{\text{best}}$ unchanged for 5 iterations **then**
14. *Stagnation Retreat*: Back-step toward $\mathbf{x}_{\text{best}}$ with cosine-geometric decay
15. **end if**
16. **end for**

3.2 SOS Mutualism for Exploration

The original CPO visual (Strategy 1) and acoustic (Strategy 2) defense strategies operate through random individual differentiation. In the spherical space, this frequently produces search directions from infeasible individuals (those with infinite cost), rendering the differential guidance ineffective. Sphere-ICPO replaces these two strategies with Symbiotic Organisms Search (SOS) mutualism, originally proposed by Cheng and Prayogo and first applied to CPO by Liu et al. in Cartesian space:

$$\mathbf{RMV} = \mathbf{x}_i + r_1 \cdot (\mathbf{x}_{\text{rand}} - \mathbf{x}_i) \quad (6)$$

$$\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + r_2 \cdot (\mathbf{x}_{\text{best}} - \mathbf{RMV}) \quad (7)$$

where $\mathbf{x}_{\text{rand}}$ is a randomly selected partner agent, $\mathbf{RMV}$ is the mutualism vector representing the collaborative midpoint, and $r_1, r_2 \in [0, 1]$ are uniform random numbers. Equation (6) establishes a cooperative relationship between the current agent and a randomly chosen partner, while Equation (7) guides both toward the global best solution. Unlike the original CPO strategies that rely on the difference between two randomly selected individuals ($\mathbf{x}_{r1} - \mathbf{x}_{r2}$), the SOS mechanism constructs a robust search direction through partner cooperation and global-best correction. This represents the first application of SOS mutualism in the spherical vector space for UAV path planning.

3.3 Adaptive Exploration Ratio

The original CPO employs a purely random switch $\text{rand}() < \text{rand}()$ to decide between exploration and exploitation, resulting in a fixed 50% exploration probability regardless of the optimization stage. This prevents the algorithm from naturally transitioning from broad exploration in early iterations to focused exploitation near convergence. Sphere-ICPO introduces an adaptive nonlinear decay function:

$$a(t) = 2 \left(0.7 \left(1 - \frac{t}{T} \right)^{0.5} + 0.3 \right), \quad \text{expl_ratio} = \frac{a(t)}{2} \quad (8)$$

where $a(t) \in [0.6, 2.0]$ mimics the exploration control parameter of GWO, and the normalized $\text{expl_ratio} \in [0.3, 1.0]$ determines the probability of invoking SOS mutualism (exploration) versus CPO exploitation strategies. At the start of optimization ($t \approx 0$), $\text{expl_ratio} \approx 1.0$, favoring broad exploration. As $t \rightarrow T$, $\text{expl_ratio} \rightarrow 0.3$, shifting the focus to local exploitation. This smooth transition enables the algorithm to explore the search space aggressively when little is known and exploit precisely when approaching convergence.

3.4 Stagnation-Triggered Periodic Retreat

The periodic retreat strategy, originally proposed by Liu et al. in ICPO, helps the algorithm escape local optima by periodically moving agents backward from the current best position using cosine-geometric calculations. However, the original implementation triggers retreat unconditionally at fixed intervals (every 10% of total iterations), which can disrupt ongoing convergence when the algorithm is making steady progress. Sphere-ICPO refines this mechanism with a stagnation-triggered activation policy.

Let $K = 5$ be the stagnation threshold. When the global best cost remains unchanged for K consecutive iterations, the retreat is activated. The retreat step is computed as:

$$r_g = 0.9 - \frac{\log(t+1) \cdot (0.9 - 0.1)}{\log(T+1)} \quad (9)$$

$$r = \text{rand}() \cdot r_g \quad (10)$$

$$\mathbf{L} = \mathbf{x}_{\text{best}} - \mathbf{x}_i \quad (11)$$

$$\alpha = \sqrt{|\mathbf{L}|^2 + |\mathbf{L}_p|^2 - 2|\mathbf{L}||\mathbf{L}_p|\cos(2\pi \cdot \text{rand}())} \quad (12)$$

$$\mathbf{x}_i^{\text{new}} = \mathbf{x}_i + r \cdot \alpha \quad (13)$$

where r_g is the logarithmically decaying retreat radius (from 0.9 to 0.1), $\mathbf{L}$ is the direction vector toward the global best, $\mathbf{L}_p$ is a randomly perturbed direction, and α is the cosine-law retreat distance. The logarithmic decay ensures large retreat steps in early stages (when escaping deep local optima is critical) and fine adjustments in later stages. The stagnation-triggered activation avoids unnecessary retreats when the algorithm is progressing steadily, reducing the variance observed in the original fixed-interval implementation.

4 Experiments

4.1 Experimental Setup

Four benchmark scenarios are constructed from real LiDAR Digital Elevation Model (DEM) data of Christmas Island, Australia. They are ordered by threat count: Map 1 contains four sparse threats, Map 2 contains five centrally clustered threats, Map 3 contains six staggered threats, and Map 4 contains seven dense threats. All scenarios share the start point (200, 100, 150), end point (800, 800, 150), terrain, cost function, and ten free spherical vectors. Threat coordinates were frozen before the formal runs; geometry checks confirmed boundary validity, start/end clearance, and the existence of a path with an 11 m safety margin.

All algorithms use the same spherical encoding, finite-solution initialization rule, population size (500), and iteration budget (200). Each algorithm is executed independently 30 times per scenario with a preregistered seed schedule. Every run stores its final path, convergence history, elapsed time, and actual number of objective evaluations. Results are reported as mean $\pm$ standard deviation. Pairwise comparisons use two-sided Wilcoxon rank-sum tests with Holm correction and Vargha–Delaney A_{12} effect sizes; overall ranks are assessed with the Friedman test.

4.2 Ablation Study

To isolate the contribution of each innovation, four variants are constructed by systematically removing components from the full Sphere-ICPO: *noSOS* replaces SOS mutualism with CPO’s original visual and acoustic strategies (while retaining adaptive exploration ratio and retreat), *noAdapt* replaces the adaptive exploration ratio with a fixed 50% probability (while retaining SOS and retreat), and *noRetreat* removes the stagnation-triggered retreat (while retaining SOS and adaptive exploration). The baseline CPO retains the original four defense strategies without any innovations.

Table 1 reports the ablation study on Map 3. Replacing SOS with the original CPO exploration operators produces the worst result (5646.27), even worse than the CPO baseline (5554.94), while all SOS-containing variants improve on CPO. This identifies SOS mutualism as the robust core change. However, the full method (5407.07) does not outperform *noAdapt* (5380.48) or *noRetreat* (5350.48) in mean cost. Consequently, this experiment does not support the stronger claim that the adaptive schedule and retreat independently improve mean performance under every setting; their role is better interpreted as conditional search-control mechanisms.

Table 1: Ablation study on Map 3 (30 independent runs, 500 particles).

Variant	SOS	Adapt. a	Retreat	Mean $\pm$ Std
CPO (baseline)	–	–	–	5554.94 $\pm$ 108.65
noSOS	–	✓	✓	5646.27 $\pm$ 68.91
noAdapt	✓	–	✓	5380.48 $\pm$ 204.31
noRetreat	✓	✓	–	5350.48 $\pm$ 165.71
Sphere-ICPO	✓	✓	✓	5407.07 $\pm$ 212.76

4.3 Multi-Map Comparison

Table 2 compares Sphere-ICPO with SPSO, GWO, CPO, and WOA. Sphere-ICPO achieves the lowest mean cost on Maps 1, 2, and 4. GWO is best on Map 3, where Sphere-ICPO ranks third behind GWO and SPSO. Across all four scenarios, Sphere-ICPO consistently improves on its direct CPO baseline; its advantage over the strongest non-CPO baselines is scenario dependent.

The lowest overall average rank belongs to Sphere-ICPO (1.50), followed by GWO (2.00), SPSO (2.50), CPO (4.00), and WOA (5.00). The narrow differences among Sphere-ICPO, SPSO, and GWO on some maps motivate the corrected pairwise analysis below.

Table 2: Multi-map comparison (30 runs; 500 particles and 200 iterations for every algorithm). Best means are bold.

Map	SPSO	GWO	CPO	WOA	Sphere-ICPO
1: 4 sparse	4705.02 $\pm$ 89.31	4691.94 $\pm$ 3.22	4775.28 $\pm$ 45.71	5610.32 $\pm$ 611.86	4686.81 $\pm$ 3.17
2: 5 clustered	5060.89 $\pm$ 11.52	5071.25 $\pm$ 10.55	5169.57 $\pm$ 29.37	5971.80 $\pm$ 501.84	5054.58 $\pm$ 4.66
3: 6 staggered	5260.68 $\pm$ 232.08	5235.05 $\pm$ 180.56	5551.59 $\pm$ 72.06	6794.28 $\pm$ 465.61	5353.75 $\pm$ 212.70
4: 7 dense	4909.26 $\pm$ 72.12	4900.53 $\pm$ 77.82	5073.04 $\pm$ 30.94	5999.41 $\pm$ 507.89	4897.10 $\pm$ 76.94

4.4 Statistical Validation

Table 3 reports Holm-corrected p -values from two-sided Wilcoxon rank-sum tests. Sphere-ICPO significantly outperforms CPO and WOA on every map. It significantly outperforms all four baselines on Map 2. On Map 1 its advantage over SPSO is not significant, although the gains over GWO, CPO, and WOA are significant. On Map 3 it is significantly worse than SPSO and GWO but better than CPO and WOA. On Map 4 the small mean advantages over SPSO and GWO are not significant.

The superscript $-$ marks comparisons in which Sphere-ICPO has the higher (worse) mean. The Friedman test yields average ranks of Sphere-ICPO (1.50), GWO (2.00), SPSO (2.50), CPO (4.00), and WOA (5.00), where lower is better. The overall difference is significant ($p = 0.00869$). Thus Sphere-ICPO achieves the best aggregate rank, while the map-level tests show that its advantage is not universal.

Table 3: Holm-corrected Wilcoxon p -values for Sphere-ICPO versus each baseline.

Map	vs. SPSO	vs. GWO	vs. CPO	vs. WOA
1: 4 sparse	0.1297	7.61×10^{-7}	1.21×10^{-10}	1.21×10^{-10}
2: 5 clustered	0.00334	6.39×10^{-9}	1.21×10^{-10}	1.21×10^{-10}
3: 6 staggered	0.0232 ⁻	0.0206 ⁻	0.00411	1.21×10^{-10}
4: 7 dense	0.9285	0.9285	5.33×10^{-10}	1.21×10^{-10}

5 Conclusion

This paper proposed Sphere-ICPO, a spherical vector-based improved Crested Porcupine Optimizer for UAV 3D path planning. In a frozen four-map benchmark with equal population and iteration budgets, Sphere-ICPO achieves the lowest mean cost on three maps and the best overall average rank (1.50); the Friedman test gives $p = 0.00869$. Holm-corrected tests confirm strong improvements over the direct CPO baseline and WOA on every map, while comparisons with SPSO and GWO remain scenario dependent. The ablation study identifies SOS mutualism as the robust core improvement, but does not show independent mean-cost benefits from the adaptive schedule and retreat on Map 3. This limitation should motivate future work on conditional activation or parameter adaptation rather than stronger universal claims. Extensions to multi-UAV planning and evaluation under explicit function-evaluation budgets are also promising directions.

References

- [1] M.D. Phung and Q.P. Ha, “Safety-enhanced UAV path planning with spherical vector-based particle swarm optimization,” *Applied Soft Computing*, vol. 107, p. 107376, 2021.
- [2] S. Mirjalili, S.M. Mirjalili, and A. Lewis, “Grey Wolf Optimizer,” *Advances in Engineering Software*, vol. 69, pp. 46–61, 2014.
- [3] S. Mirjalili and A. Lewis, “The Whale Optimization Algorithm,” *Advances in Engineering Software*, vol. 95, pp. 51–67, 2016.
- [4] M. Abdel-Basset, R. Mohamed, and M. Abouhawwash, “Crested Porcupine Optimizer: A new nature-inspired metaheuristic,” *Knowledge-Based Systems*, vol. 284, p. 111257, 2024.
- [5] S. Liu, Z. Jin, H. Lin, and H. Lu, “An improve crested porcupine algorithm for UAV delivery path planning in challenging environments,” *Scientific Reports*, vol. 14, p. 20445, 2024.

- [6] J. Liu et al., “A Q-Learning Crested Porcupine Optimizer for Adaptive UAV Path Planning,” *Machines*, vol. 13, p. 566, 2025.
- [7] H. Zhang et al., “Application of Improved Crown Porcupine Optimizer in UAV Path Planning Based on Dynamic Weighted JAYA-CPO Attack Strategy,” *Protection and Control of Modern Power Systems*, vol. 10, no. 6, 2025.
- [8] X. Zhang, Z. Cheng, and Z. Zheng, “Multi-Strategy optimisation algorithm for multi-UAV path planning in complex mountainous environments,” *Scientific Reports*, 2026.
- [9] Z. Shi and C. Li, “A Reference-Point Guided Multi-Objective Crested Porcupine Optimizer for Global Optimization and UAV Path Planning,” *Mathematics*, vol. 14, p. 380, 2026.

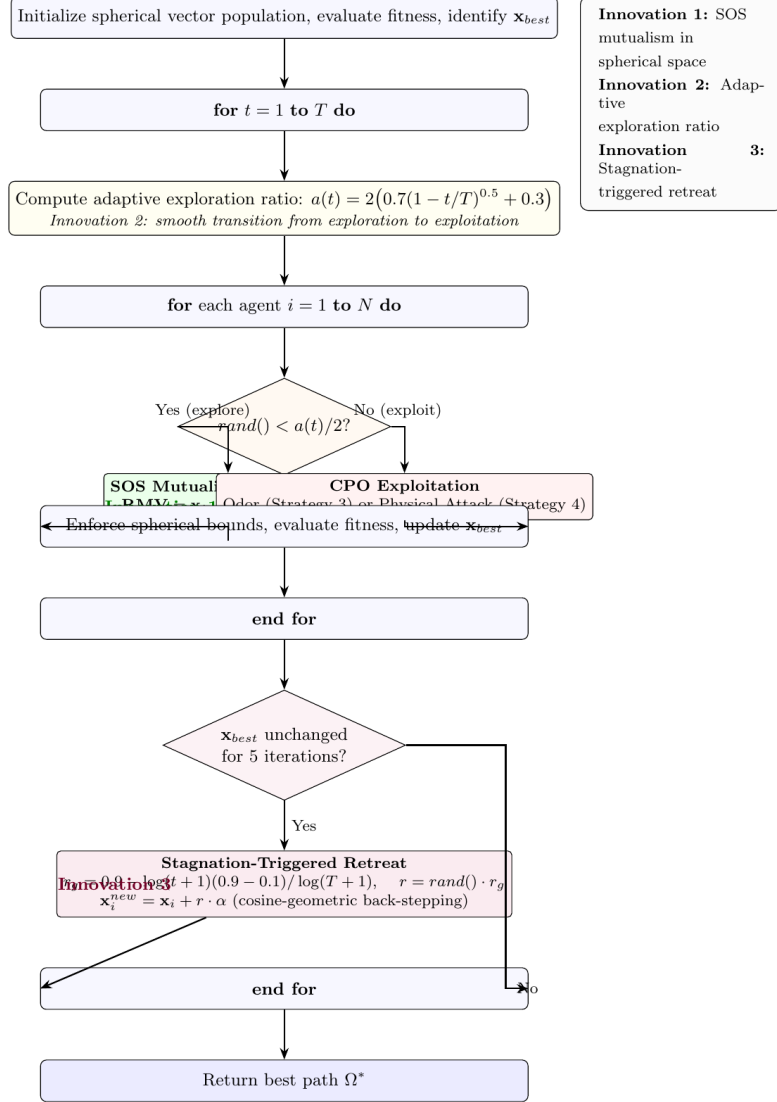


Figure 2: Flowchart of the Sphere-ICPO algorithm. Three innovations are highlighted: (1) SOS mutualism replaces CPO’s visual and acoustic strategies for robust exploration in spherical space; (2) an adaptive exploration ratio $a(t)$ enables smooth transition from exploration to exploitation; (3) stagnation-triggered retreat provides targeted local-optimum escape.

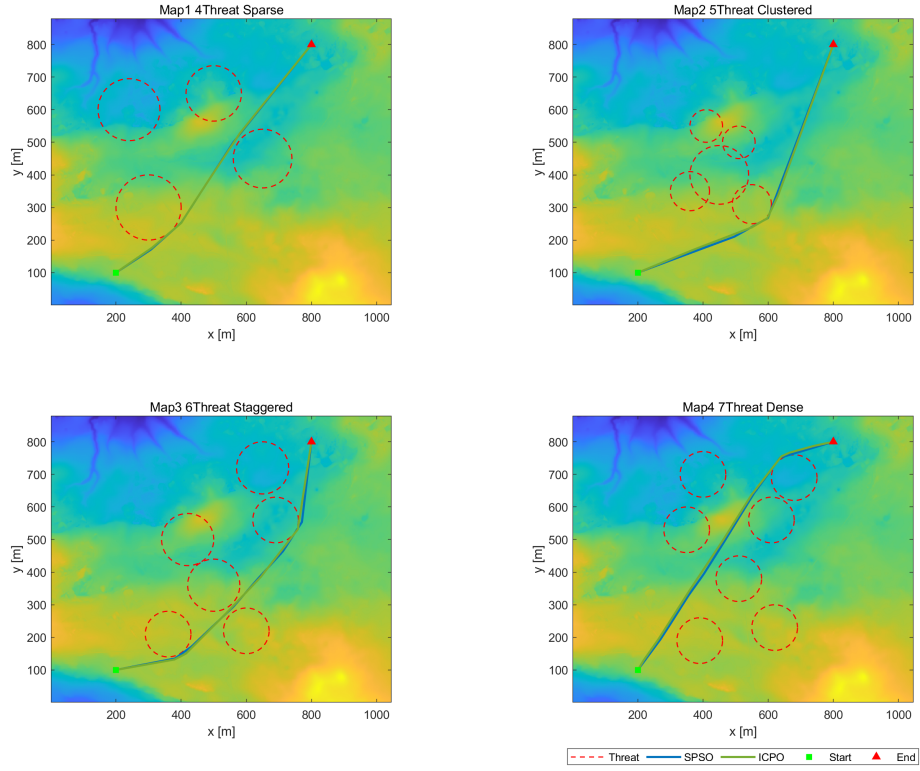


Figure 3: Frozen benchmark scenarios and the best paths found by SPSO and Sphere-ICPO. Threats are red dashed circles; all maps share the same start and end points.

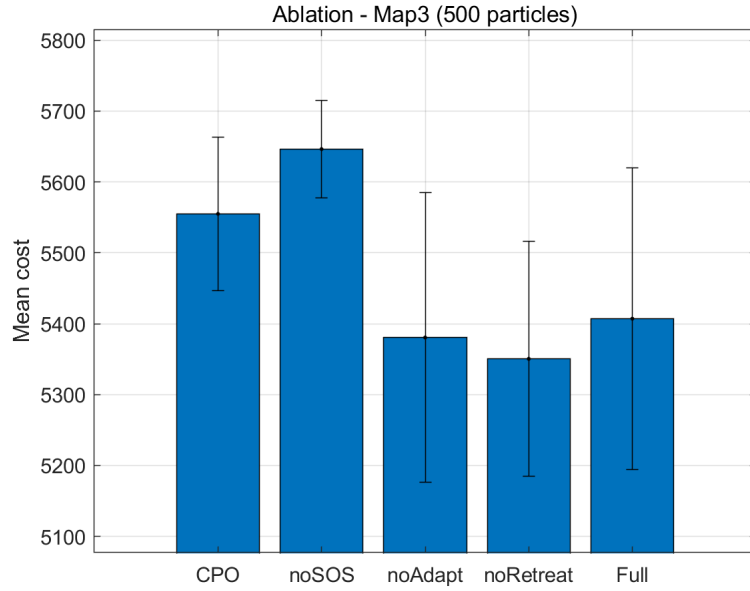


Figure 4: Map 3 ablation means with standard-deviation error bars.

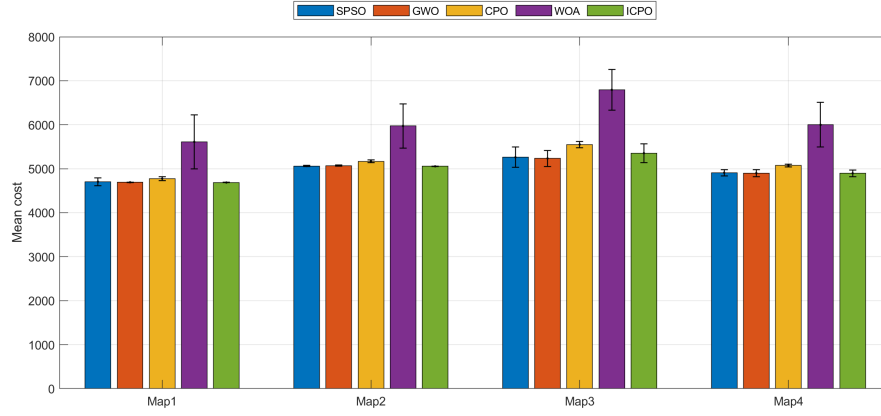


Figure 5: Mean costs with standard-deviation error bars across the frozen benchmark.

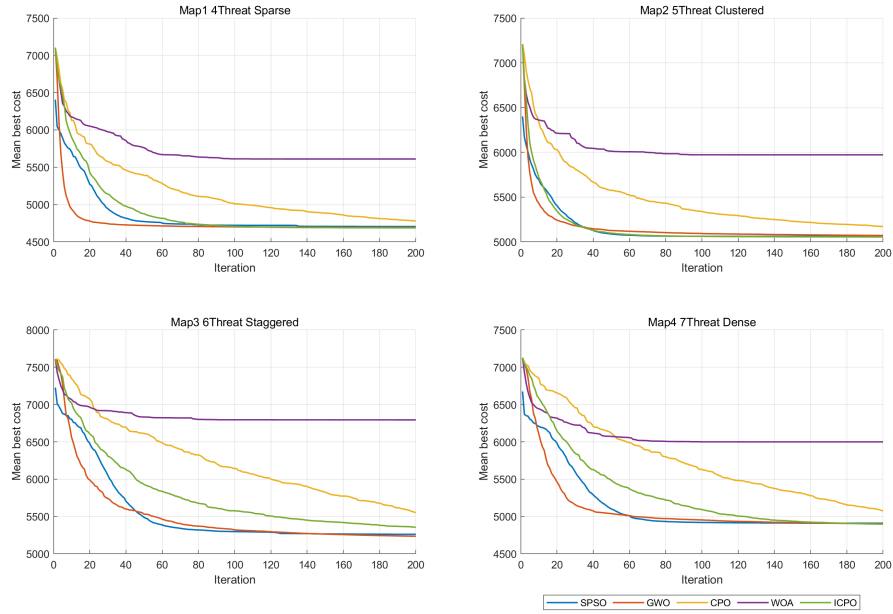


Figure 6: Mean best-so-far convergence curves over 30 independent runs.

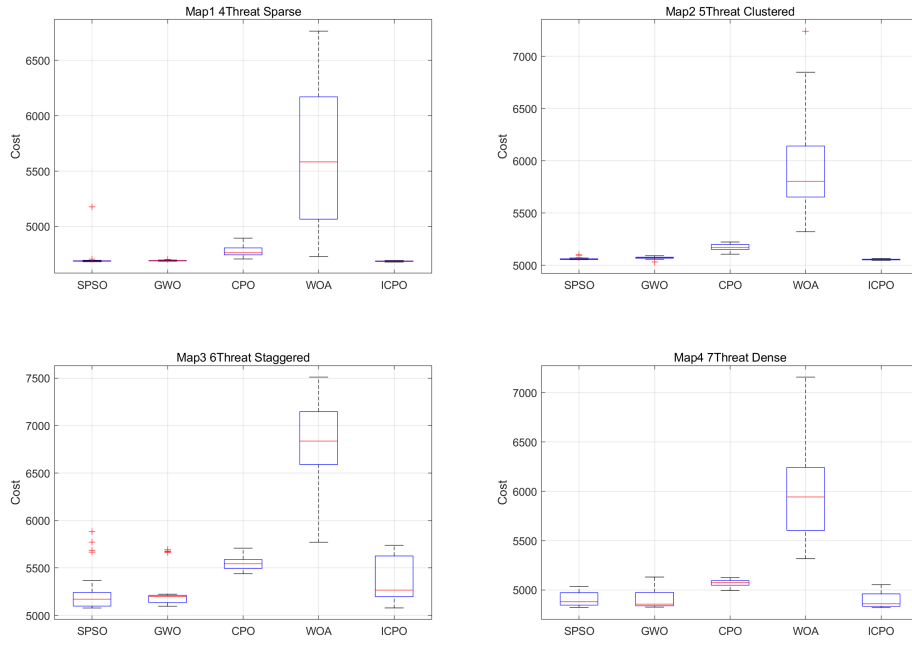


Figure 7: Cost distributions across four maps and five algorithms (30 runs each).